

Fixture: plan_struct_causes / golden-bad-orphan

Identical to the golden-good fixture except RC-03's LAST column ("Removed / measured by") cell is EMPTY -- the row names no task at all. Per research.md's own §2 preamble (lines 164-166 of the real document): "Task ids in the last column refer to the phased plan in plan.md ... Every row is removed or measured by at least one task ... (no orphan cause, no orphan task -- checked mechanically by T-H08)." A row whose last cell is blank is an orphan cause at the register level -- detectable from research.md alone (no cross-reference to plan.md's own task list is needed for THIS check; that deeper bipartite cross-check against plan.md's actual tasks is SC-C-003, the separate `plan` subcommand). Expected `causes` verdict: exit 1, naming RC-03.

- Root-cause register

- Register table

RC	Cause	Origin	Class	Measured magnitude (named denominator)	Share of total cycle	Evidence	Settling evidence / what remains	Removed / measured by
RC-01	Fixture cause one	O1	CONFIRMED	1 of 1 fixture measurement	UN-MEASURED	fixture evidence line one	settling note (T-X01)	T-X01
RC-02	Fixture cause two	O2	CONFIRMED	1 of 1 fixture measurement	UN-MEASURED	fixture evidence line two	settling note (T-X02)	T-X02
RC-03	Fixture cause three	O3	REFUTED	fixture refutation note	n/a	fixture evidence line three	n/a	
RC-04	Fixture cause four	O4	UNDETERMINED	fixture magnitude unmeasured	UN-MEASURED	fixture evidence line four	settling evidence needed (T-X04)	T-X04

. Register counts (machine count over the
Class column of § .)

Class	Rows	Row ids
CONFIRMED	2	RC-01, RC-02
REFUTED	1	RC-03
UNDETERMINED	1	RC-04
Total	4	